# REQUIRMENT PHASE

## 1.Introduction:

The Polyline Editor Application is a graphical interactive system that allows users to create and manipulate polylines (connected line segments).
The purpose of this system is to provide a simple, efficient, and user-friendly interface for drawing and editing shapes using points.
This requirements phase defines:
What the system should do
How users will interact with it
What constraints and improvements are needed

## 2. Target Users:

- The intended users include:
- Students learning computer graphics
- Beginner users practicing drawing tools
- Users needing simple shape editing

**User Characteristics:**

- Basic computer knowledge
- Ability to use mouse and keyboard
- No prior experience required

## 3. Current Situation:

Currently, users rely on general tools like Paint applications. However:

- No proper point-based editing system
- Difficult to adjust shapes precisely
- Lack of direct manipulation features

Therefore, a dedicated polyline editor is required to improve usability and precision.

## 4. Functional Requirements:

**1. Create Polyline**
User shall start drawing using key 'b'
System shall initiate a new polyline
**2. Add Points**
Each mouse click shall add a new point
Points shall automatically connect
A line shall appear only after two points
**3. Move Points**
User shall press 'm' to activate move mode
Selected point shall be highlighted
User shall drag the point
System shall update connected lines in real-time
**4. Delete Points**
User shall press 'd'
Select a point to remove
System shall reconnect remaining points automatically
**5. Refresh Screen**
User shall press 'r'
System shall redraw the canvas
**6. Manage Multiple Polylines**
User shall create multiple polylines
Each polyline shall behave independently

**7. Exit Application**
User shall press ‘q’ to exit

## ***5. Non-Functional Requirements:***

- Usability

Interface should be simple and intuitive
Commands should be easy to remember
Minimal user effort required

- Performance

System should respond instantly
Smooth dragging and interaction

- Reliability

No crashes during normal use
All actions should behave consistently

- Learnability

Beginners should learn quickly
On-screen guidance should be available

- Error Handling (from analysis)

Invalid clicks should be ignored
No unnecessary error popups
System should prevent incorrect actions

## ***6. User Environment:***

- Desktop computer
- Mouse (for drawing & selecting points)
- Keyboard (for commands: b, m, d, r, q)
- Graphical display interface

## ***7. Assumptions:***

- User understands basic mouse interaction
- System operates in a 2D environment
- Only simple polylines are required

## ***8. Constraints:***

- Limited development time
- Simple UI (no complex graphics)
- Basic input methods only

## ***9. Problem-Based Requirements:***

- Identified Problem
- Requirement Solution
- Wrong point selection
- Highlight selected point
- Confusion in controls
- Display instructions on screen
- Shape distortion after delete
- Smooth automatic reconnection
- Lack of feedback
- Real-time visual updates
- User errors
- Ignore invalid clicks

## ***10. HCI Principles Applied:***

Based on graphical analysis:

- Efficiency

"Move Point" is most frequent → must be optimized

- Usability

Majority actions are valid → system should remain simple

- Error Prevention

Invalid inputs should be ignored

- Feedback

Visual changes must appear immediately

## ***11. Conceptual Model:***

The system follows Direct Manipulation:

- Users interact directly with points
- Immediate visual response
- Real-time updates

This improves user control and understanding

## ***12. Visual Feedback Requirements:***

System must:

- Highlight selected points
- Update lines instantly
- Avoid unnecessary alerts
- Provide smooth interaction

## ***Final Conclusion:***

The Polyline Editor application is designed to provide a simple, interactive, and efficient drawing environment.

By focusing on:

- Usability
- Direct manipulation
- Error handling
- The system ensures a smooth and user-friendly experience for all types of users.

# ***Requirement Prioritization Graph:***

```
Importance (High)
    ↑
    |
    |   ● Move Points (m)
    |   ● Add Points
    |   ● Create Polyline (b)
    |   ● Real-time Feedback
    |   ● Highlight Point
    |
    |   ● Delete Points (d)
    |   ● Auto Reconnection
    |   ● Error Handling
    |
    |
    |   ● Refresh Screen (r)
    |
```

```
|     ● Exit Application (q)
|     ● On-screen Instructions
|
+-------------------------------------→ Frequency (High)
Low                          High
```